Likelihood:

$$y_{i,t} \sim \mathcal{N}\left(\mu_{i,t} + \epsilon_{i,t},\ \tau_i^{-1}\right), \quad t > 1$$

Where y is the raw weekly rate, i is the health region, and t is week

Mean structure:

$$\begin{aligned}
\mu_{i,t} = \alpha_i & & \leftarrow \quad \textbf{Baseline} \\
&+ \beta_i \cos\left(\tfrac{2\pi t}{52}\right) + \gamma_i \sin\left(\tfrac{2\pi t}{52}\right) & \leftarrow \quad \textbf{Annual cycle} \\
&+ \delta_{i,t}\, \mathbf{1}(t = \text{New Year week} - 2) + \delta_{i,t}\, \mathbf{1}(t = \text{New Year week} - 1) & \leftarrow \quad \textbf{New Year effect} \\
&\qquad + \delta_{i,t}\, \mathbf{1}(t = \text{New Year week}) & \\
&\qquad + \delta_{i,t}\, \mathbf{1}(t = \text{New Year week} + 1) & \\
&+ \kappa_{i=\text{MW},\,t}\, \mathbf{1}(t = \text{Aug 8 2024 week}) + \kappa_{i=\text{MW},\,t}\, \mathbf{1}(t = \text{Aug 15 2024 week}) & \leftarrow \quad \textbf{Mid West reset} \\
&\qquad + \kappa_{i=\text{MW},\,t}\, \mathbf{1}(t = \text{Aug 22 2024 week}) & \\
&\qquad + \kappa_{i=\text{MW},\,t}\, \mathbf{1}(t = \text{Aug 29 2024 week}) & \\
&\qquad + \kappa_{i=\text{MW},\,t}\, \mathbf{1}(t = \text{Sep 5 2024 week}) &
\end{aligned}$$

AR(1) Component:

$$\epsilon_{i,t} = \phi\left(y_{i,t-1} - \mu_{i,t-1}\right)$$

Uninformative priors:

$$\alpha_i,\ \beta_i,\ \gamma_i,\ \delta_{i,t},\ \kappa_{i=\text{MW},\,t} \sim \mathcal{N}(0,\ 0.001)$$

$$\tau_i \sim \Gamma(0.001,\ 0.001) \qquad \phi \sim \text{Uniform}(-1,\ 1)$$

Normal priors $\mathcal{N}(\mu, \tau)$ use precision in the 2nd argument.